USER MANUAL · REV 1.0

OCD

Quad-Preset MIDI Drive · a faithful clone of the OCD-style overdrive · handmade in Austria by Walter Craffonara

NOTE This pedal is a faithful, circuit-for-circuit clone of the OCD-style overdrive, expanded with functions I added: a relay-switched quad-preset system and MIDI. The name “OCD” is used only to identify the original circuit this build recreates. CRAFFONARA FX is an independent builder and is not affiliated with, authorised by or endorsed by Fulltone Musical Products, Inc.

1 Overview & Concept

At its core this is a faithful clone of the OCD-style overdrive — the same circuit, the same voicing. What I added is the part the original never had: the gain-shaping controls are doubled and relay-switched, so you carry four complete settings at once and jump between them by PRESET button or over MIDI.

The DRV and TONE controls each have two pots — one white, one red. A relay picks which pot is live per channel, and the two relays work independently, so nothing is in the audio path except the selected pot. VOL sets the overall output. The CLIP and FILTER toggles and a set of internal DIP switches let you dial in any of the historic OCD-style versions.

2 Controls & Connections

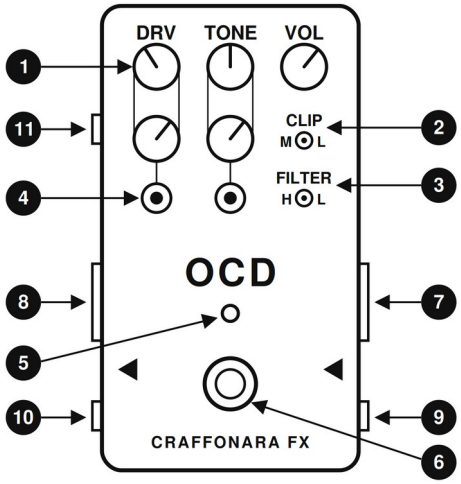


Fig. 1 — Panel and connections

1 DRV · TONE · VOL the main controls for drive, tone colour and output level. DRV and TONE each exist twice: an upper pot with a white marker and a lower pot with a red marker.

2 CLIP toggle M = MOSFET / germanium clipping character, L = LED clipping for a louder, tighter, more aggressive edge.

3 FILTER toggle H = high peak, brighter and more cutting; L = low peak, rounder and darker.

4 PRESET buttons one soft-touch button per channel (DRV, TONE), each driving its own relay. LED ring dark = preset #1 (upper pot, white marker); LED ring lit = preset #2 (lower pot, red marker).

5 STATUS LED shows whether the effect is engaged or bypassed. It does NOT indicate which preset is active — that is shown by the LED ring around each PRESET button.

6 FOOTSWITCH silent soft-touch switch, engages / bypasses the effect (relay true bypass).

7 INPUT 6.35 mm mono jack from your guitar or the previous pedal.

8 OUTPUT 6.35 mm mono jack to your amp or the next pedal.

9 MIDI IN TRS socket for incoming MIDI (Type A).

10 MIDI THRU TRS socket passing incoming MIDI on to the next device.

11 POWER 9–18 V DC, center-negative — 18 V gives more headroom and dynamics.

3 The Quad-Preset System

DRV and TONE each exist twice on the panel: an upper pot with a white marker and a lower pot with a red marker. Each channel has its own PRESET button (4), and each button drives its own relay — the two relays work completely independently of one another.

The LED ring around a PRESET button shows which pot that channel is using:

LED ring dark (preset #1) the upper pot with the white marker is active; the lower pot is muted.

LED ring lit (preset #2) the lower pot with the red marker is active; the upper pot is muted.

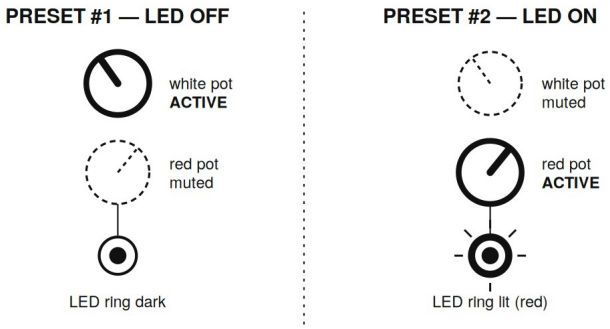


Fig. 2 — Preset states of one channel (DRV or TONE)

Because the two relays switch independently, the two channels give you 2 × 2 = four complete settings in one pedal — a quad-preset system. Recall any of the four by PRESET button or over MIDI: for example a rhythm sound, a lead sound, and two in-between voicings.

> **NOTE** Preset switching is relay-based and click-free. Only the selected pot sits in the signal path, so the muted pot causes no tone loss.

4 Voicing Controls

CLIP toggle — M / L M gives the classic MOSFET / germanium clipping of the OCD-style circuit; L switches to LED clipping for a louder, tighter, more aggressive edge. The toggle only works with the matching +GE DIP setting — see section 5.

FILTER toggle — H / L sets the tone-stack peak. H is brighter and cuts through a dense mix; L is rounder and darker.

5 OCD-Style Versions (internal DIP)

Six internal DIP switches recreate the voicing changes across the OCD-style production run, so one pedal covers the whole family from the early V1.x boards to the later V2 germanium. Set the switches as shown below to reach the version you want. The pedal ships in the V1.7 setting.

Version	+GE (2 switches)	HI CUT	+BASS	+MIDS left	+MIDS right
V1.1 / V1.2	**both ON**	OFF	**ON**	**ON**	**ON**
V1.3	**both ON**	OFF	**ON**	OFF	**ON**
V1.4	one ON, one OFF	OFF	OFF	OFF	**ON**
V1.5	**both ON**	**ON**	OFF	OFF	**ON**
V1.6 / V1.7 (default)	**both ON**	OFF	OFF	OFF	**ON**
V2.0	**both ON**	OFF	OFF	OFF	OFF
V2 Custom Shop Germanium	both OFF	OFF	OFF	OFF	OFF

The left +MIDS switch needs the right +MIDS switch ON to give the V1.1 / V1.2 voicing. Left ON with right OFF matches no historic version, but is a useful in-between mid voicing.

> **NOTE** The CLIP toggle (2) only has an effect when at least one +GE switch is OFF (the V1.4 and V2 Custom Shop Germanium settings). With both +GE switches ON the clipping is MOSFET-only and the CLIP toggle changes nothing.

6 MIDI Reference

The pedal responds to MIDI Control Change (CC) messages on its MIDI channel (see the next section).

CC	Value	Command
00	00	TONE preset #1 and DRV preset #1
	01	TONE preset #2 and DRV preset #1
	02	TONE preset #1 and DRV preset #2
	03	TONE preset #2 and DRV preset #2
10	00	TONE preset #1
	01	TONE preset #2
	02	Toggle TONE preset #1 / #2
20	00	STATE: OFF — effect bypassed
	01	STATE: ON — effect engaged
30	00	DRV preset #1
	01	DRV preset #2
	02	Toggle DRV preset #1 / #2

7 Setting the MIDI Channel

1. Disconnect the pedal from the power supply.
2. Press and hold the FOOTSWITCH while connecting the power supply — the STATUS LED lights up and starts to flash, showing the pedal is in channel-set mode.

3. Press the FOOTSWITCH as many times as the channel number you want to set (e.g. three times for MIDI channel 3). The STATUS LED flashes back with each press.
4. Press and hold the FOOTSWITCH until the STATUS LED goes dark and the pedal switches off completely.
5. On the next power-up the pedal responds to the new MIDI channel.

8 Power

Run the pedal on 9–18 V DC, center-negative, from any regulated pedalboard supply. There is no battery compartment. At 18 V the circuit has noticeably more headroom and dynamic range — a slightly louder, more open feel.

9 Specifications

Parameter	Value
Type	Overdrive — faithful clone of the OCD-style circuit, with a quad-preset system and MIDI
Controls	DRV ×2, TONE ×2 (relay-switched presets), VOL, CLIP toggle, FILTER toggle
Presets	2 × DRV, 2 × TONE — two independent relays, four combinations
Bypass	Relay true bypass, silent soft-touch footswitch
MIDI	IN + THRU (TRS, Type A), Control Change
Internal options	6 DIP switches: +GE ×2, HI CUT, +BASS, +MIDS ×2 (V1.1–V1.7 and V2 voicings)
Audio I/O	6.35 mm mono input and output
Power	9–18 V DC, center-negative
Enclosure	Brushed aluminium, hand-finished

NOTE Trademark notice: “OCD” and “Fulltone” are trademarks of Fulltone Musical Products, Inc. They are used here purely to describe the original circuit that this independent build recreates. CRAFFONARA FX is not affiliated with, authorised by or endorsed by Fulltone.